Section 3.5: Spanning Sets and Linear Independence

(Reference: Section 2.3)

1) A vector $\vec{\mathbf{w}}$ is called a <u>linear combination</u> of the vectors $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r$ if it can be expressed in the form

$$\vec{\mathbf{w}} = k_1\vec{\mathbf{v}}_1 + k_2\vec{\mathbf{v}}_2 + \cdots + k_r\vec{\mathbf{v}}_r$$

where $k_1, k_2, \cdots, k_r$ are scalars.

Note: We can always solve a vector equation (in general, a system of linear equations) by using row operations on the augmented matrix using the vectors as columns on the left-hand side of the separator.

2) If $S = \{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r\}$ are vectors in $\mathbb{R}^n$, then the set W of all linear combinations of $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r$ is called the <u>span</u> of $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r$ denoted by

$$\mathrm{W} = \mathrm{span(S)} \quad \text{or} \quad W = \mathrm{span}\{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r\}$$

and we say that the vectors $\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r$ <u>span W</u>.
If $\mathrm{span(S)} = \mathbb{R}^n$, then S is called a <u>spanning set for $\mathbb{R}^n$</u>.

Note: The linear system $A\vec{\mathbf{x}} = \vec{\mathbf{b}}$ is consistent if and only $\vec{\mathbf{b}}$ is in the span of the columns (that is, column vectors) of A.

Example 3.5.1. Show that $\mathbb{R}^3 = \mathrm{span}\left(\begin{bmatrix}3\\4\\1\end{bmatrix}, \begin{bmatrix}4\\3\\1\end{bmatrix}, \begin{bmatrix}3\\3\\1\end{bmatrix}\right)$.

Solution: (We need to show that any vector $\begin{bmatrix}x\\y\\z\end{bmatrix} \in R^3$ can be expressed as a linear combination of the three vectors.)

$$k_1\begin{bmatrix}3\\4\\1\end{bmatrix} + k_2\begin{bmatrix}4\\3\\1\end{bmatrix} + k_3\begin{bmatrix}3\\3\\1\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix} \Longrightarrow \begin{bmatrix}3k_1 + 4k_2 + 3k_3\\4k_1 + 3k_2 + 3k_3\\k_1 + k_2 + k_3\end{bmatrix} = \begin{bmatrix}x\\y\\z\end{bmatrix}$$

$$\left[\begin{array}{ccc|c}3 & 4 & 3 & x\\4 & 3 & 3 & y\\1 & 1 & 1 & z\end{array}\right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c}1 & 1 & 1 & z\\4 & 3 & 3 & y\\3 & 4 & 3 & x\end{array}\right] \xrightarrow{R_2+(-4)R_1, R_3+(-3)R_1} \left[\begin{array}{ccc|c}1 & 1 & 1 & z\\0 & -1 & -1 & y-4z\\0 & 1 & 0 & x-3z\end{array}\right] \rightarrow$$

$$\xrightarrow{(-1)R_2} \left[\begin{array}{ccc|c}1 & 1 & 1 & z\\0 & 1 & 1 & 4z-y\\0 & 1 & 0 & x-3z\end{array}\right] \xrightarrow{R_3+(-1)R_2} \left[\begin{array}{ccc|c}1 & 1 & 1 & z\\0 & 1 & 1 & 4z-y\\0 & 0 & -1 & x+y-7z\end{array}\right] \rightarrow$$

$$\xrightarrow{(-1)R_3} \left[\begin{array}{ccc|c}1 & 1 & 1 & z\\0 & 1 & 1 & 4z-y\\0 & 0 & 1 & 7z-x-y\end{array}\right] \xrightarrow{R_1+(-1)R_3, R_2+(-1)R_3} \left[\begin{array}{ccc|c}1 & 1 & 0 & x+y-6z\\0 & 1 & 0 & x-3z\\0 & 0 & 1 & 7z-x-y\end{array}\right] \rightarrow$$

$$\xrightarrow{R_1+(-1)R_2} \left[\begin{array}{ccc|c}1 & 0 & 0 & y-3z\\0 & 1 & 0 & x-3z\\0 & 0 & 1 & 7z-x-y\end{array}\right] \Longrightarrow k_1 = y-3z, k_2 = x-3z, k_3 = 7z-x-y$$

Hence any vector in R^3 can be expressed as a linear combination of the three vectors $\begin{bmatrix}3\\4\\1\end{bmatrix}, \begin{bmatrix}4\\3\\1\end{bmatrix}, \begin{bmatrix}3\\3\\1\end{bmatrix}$, in other words, $\mathbb{R}^3 = \mathrm{span}\left(\begin{bmatrix}3\\4\\1\end{bmatrix}, \begin{bmatrix}4\\3\\1\end{bmatrix}, \begin{bmatrix}3\\3\\1\end{bmatrix}\right)$.

3) If $S = \{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2, \cdots, \vec{\mathbf{v}}_r\}$ is a nonempty set of vectors, then the vector equation

$$k_1\vec{\mathbf{v}}_1 + k_2\vec{\mathbf{v}}_2 + \cdots + k_r\vec{\mathbf{v}}_r = \mathbf{0}$$

has at least one solution, namely

$$k_1 = 0, k_2 = 0, \cdots, k_r = 0.$$

If this "trivial solution" is the only solution, then S is called a <u>linearly independent set</u>; if there are other solutions, then S is called a <u>linearly dependent set</u>, and any non-trivial solution corresponds to a <u>dependence relationship</u> among the vectors.

Note: A set of vectors S is linearly dependent if (and only if) at least one of the vectors in S can be expressed as a linear combination of the remaining vectors.

Dependence Relationship

4) How to determine if a set of vectors is linearly dependent?

Let A be an $m \times n$ matrix with column vectors $\vec{\mathbf{c}}_1, \vec{\mathbf{c}}_2, \cdots, \vec{\mathbf{c}}_n$ in $\mathbb{R}^m$.

Then $\vec{\mathbf{c}}_1, \vec{\mathbf{c}}_2, \cdots, \vec{\mathbf{c}}_n$ are linearly dependent if and only if the homogeneous linear system with augmented matrix $[A|\vec{\mathbf{0}}]$ has a nontrivial solution. We can determine whether this is true by using elementary row operations to convert $[A|\vec{\mathbf{0}}]$ to an upper triangular matrix.

Example 3.5.2. Use the definition of linear independence to prove that $\vec{\mathbf{v}}_1 = [1 \quad 1 \quad 2]$, $\vec{\mathbf{v}}_2 = [2 \quad 3 \quad 4]$, $\vec{\mathbf{v}}_3 = [4 \quad 1 \quad 2]$ is a linearly independent set of $\mathbb{R}^3$.

Solution: $k_1[1 \quad 1 \quad 2] + k_2[2 \quad 3 \quad 4] + k_3[4 \quad 1 \quad 2] = [0 \quad 0 \quad 0]$

$\Longrightarrow [k_1 + 2k_2 + 4k_3, k_1 + 3k_2 + k_3, 2k_1 + 4k_2 + 2k_3] = [0, 0, 0]$

$$\left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 1 & 3 & 1 & 0 \\ 2 & 4 & 2 & 0 \end{array}\right] \xrightarrow{R_2+(-1)R_1} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 1 & -3 & 0 \\ 2 & 4 & 2 & 0 \end{array}\right] \xrightarrow{R_3+(-2)R_1} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -6 & 0 \end{array}\right] \xrightarrow{\left(-\frac{1}{6}\right)R_3} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \rightarrow$$

$$\xrightarrow{R_2+(3)R_3} \left[\begin{array}{ccc|c} 1 & 2 & 4 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_1+(-4)R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right] \xrightarrow{R_1+(-2)R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$$

$\Rightarrow k_1 = k_2 = k_3 = 0$

Hence $\{[1 \quad 1 \quad 2], [2 \quad 3 \quad 4], [4 \quad 1 \quad 2]\}$ is linearly independent.

5) Some observations on linearly dependence/independence

a) A finite set of vectors that contains the zero vector is linearly dependent.

b) A set of vectors is linear dependent if and only if at least one of the vectors is a linear combination of the other vectors (that is, at least one of the vectors is in the span of the other vectors).

c) A set with exactly two vectors is linear dependent if and only if one of the vectors is a scalar multiple of the other. (Hence, geometrically, a set of two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ is linearly dependent if and only if the vectors lie on the same line when they are placed with their initial points at the origin.)

Practice Exercises

1. Find the span of the set containing the given vectors.

Keep your work and results for the next exercise.

(a) $\begin{bmatrix}-1\\3\end{bmatrix}, \begin{bmatrix}-4\\13\end{bmatrix}$ (b) $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\4\end{bmatrix}$ (c) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}$

(d) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\4\\6\end{bmatrix}$ (e) (Assignment 2) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\4\\6\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}$ (f) $\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\3\\6\end{bmatrix}$

[Answers: (a) $\left[\begin{array}{cc|c}-1 & -4 & x\\ 3 & 13 & y\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c}1 & 0 & -13x-4y\\ 0 & 1 & 3x+y\end{array}\right]$; the span is the whole space $\mathbb{R}^2$.

(b) $\left[\begin{array}{cc|c}1 & 2 & x\\ 2 & 4 & y\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c}1 & 2 & x\\ 0 & 0 & y-2x\end{array}\right]$; the span is the line $y-2x=0$.

(c) $\left[\begin{array}{ccc|c}1 & 4 & x\\ 2 & 5 & y\\ 3 & 6 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c}1 & 0 & (4/3)y-(5/3)x\\ 0 & 1 & (2/3)x-(1/3)y\\ 0 & 0 & x-2y+z\end{array}\right]$; the span is the plane $x-2y+z=0$.

(fd $\left[\begin{array}{cc|c}1 & 2 & x\\ 2 & 4 & y\\ 3 & 6 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c}1 & 2 & x\\ 0 & 0 & y-2x\\ 0 & 0 & z-3x\end{array}\right]$; the span is the line $\left\{\begin{array}{l}y-2x=0\\ z-3x=0\end{array}\right\}$.

(e) $\left[\begin{array}{ccc|c}1 & 2 & 4 & x\\ 2 & 4 & 5 & y\\ 3 & 6 & 6 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c}1 & 2 & 0 & (4/3)y-(5/3)x\\ 0 & 0 & 1 & (2/3)x-(1/3)y\\ 0 & 0 & 0 & x-2y+z\end{array}\right]$; the span is the plane $x-2y+z=0$.

(f) $\left[\begin{array}{ccc|c}1 & 1 & 1 & x\\ 1 & 2 & 3 & y\\ 1 & 3 & 6 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c}1 & 0 & 0 & 3x-3y+z\\ 0 & 1 & 0 & 5y-3x-2z\\ 0 & 0 & 1 & x-2y+z\end{array}\right]$; the span is the whole space $\mathbb{R}^3$.]

2. Use the definition to determine whether the set containing the given vectors is linearly independent or linearly dependent. You can use your results in the previous exercise.

(a) $\begin{bmatrix}-1\\3\end{bmatrix}, \begin{bmatrix}-4\\13\end{bmatrix}$ (b) $\begin{bmatrix}1\\2\end{bmatrix}, \begin{bmatrix}2\\4\end{bmatrix}$ (c) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}$

(d) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\4\\6\end{bmatrix}$ (e) $\begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}2\\4\\6\end{bmatrix}, \begin{bmatrix}4\\5\\6\end{bmatrix}$ (f) $\begin{bmatrix}1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\2\\3\end{bmatrix}, \begin{bmatrix}1\\3\\6\end{bmatrix}$

[Answers: (a) The set is linearly independent

(b) The set is linearly dependent

(c) The set is linearly independent

(d) The set is linearly dependent

(e) The set is linearly dependent

(f) The set is linearly independent]

3. Show that span$\{\vec{\mathbf{v}}_1, \vec{\mathbf{v}}_2\}$ = span$\{\vec{\mathbf{v}}_1 + \vec{\mathbf{v}}_2, \vec{\mathbf{v}}_1 - \vec{\mathbf{v}}_2\}$.

4. Given $\{\vec{\mathbf{u}}_1, \vec{\mathbf{u}}_2, \vec{\mathbf{u}}_3\}$ is a linearly independent set in $\mathbb{R}^4$, determine whether the following set is linearly independent or linearly dependent.

(a) $\{\vec{\mathbf{u}}_1 - \vec{\mathbf{u}}_2, \vec{\mathbf{u}}_2 - \vec{\mathbf{u}}_3,\}$ (b) $\{\vec{\mathbf{u}}_1 - \vec{\mathbf{u}}_2, \vec{\mathbf{u}}_2 - \vec{\mathbf{u}}_3, \vec{\mathbf{u}}_3 - \vec{\mathbf{u}}_1\}$

[Answers: (a) $\left[\begin{array}{cc|c}1 & 0 & x\\-1 & 1 & y\\0 & -1 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{cc|c}1 & 0 & x\\0 & 1 & x+y\\0 & 0 & x+y+z\end{array}\right]$; the set is linearly independent

(b) $\left[\begin{array}{ccc|c}1 & 0 & -1 & x\\-1 & 1 & 0 & y\\0 & -1 & 1 & z\end{array}\right] \xrightarrow{\text{rref}} \left[\begin{array}{ccc|c}1 & 0 & -1 & x\\0 & 1 & -1 & x+y\\0 & 0 & 0 & x+y+z\end{array}\right]$; the set is linearly dependent]

5. (Assignment 2) Find the value(s) of t so that the vectors $\vec{\mathbf{v}}_1 = [t \quad -1 \quad -1]$, $\vec{\mathbf{v}}_2 = [-1 \quad t \quad -1]$, and $\vec{\mathbf{v}}_3 = [-1 \quad -1 \quad t]$ form a linearly dependent set.

[Answer: -1 and 2]